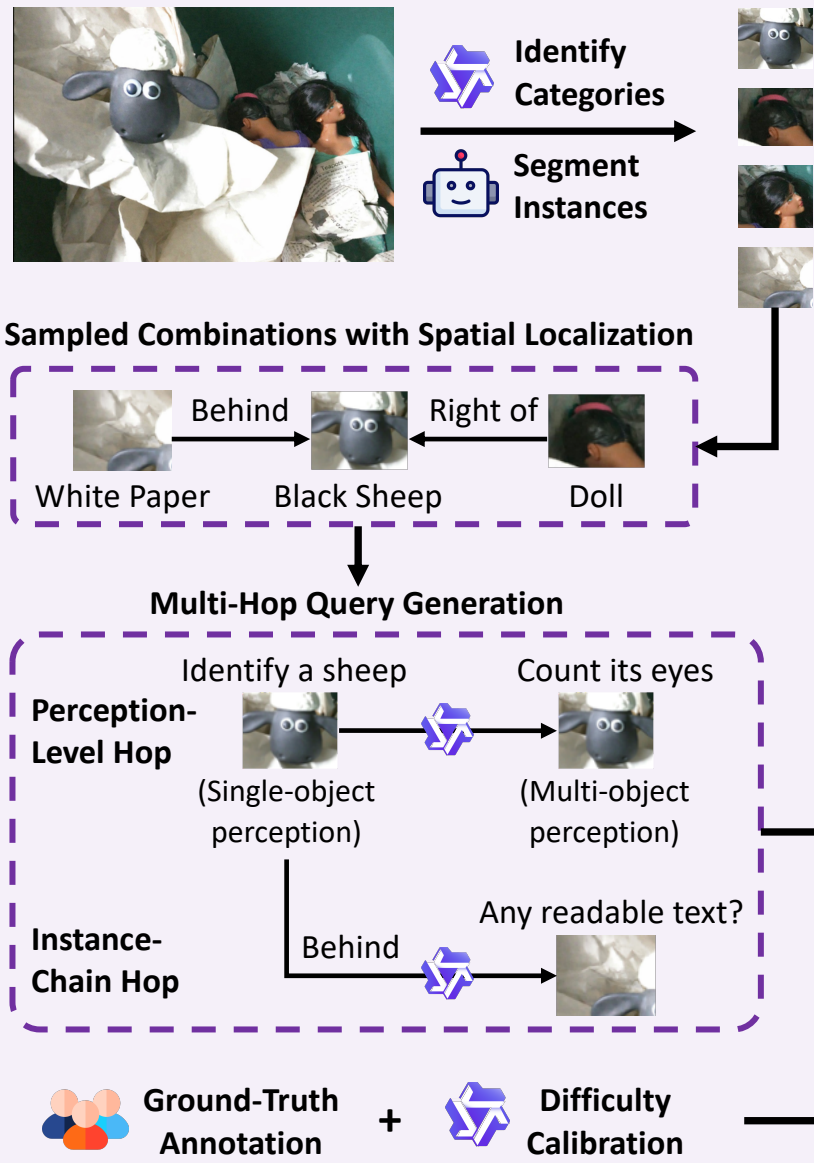
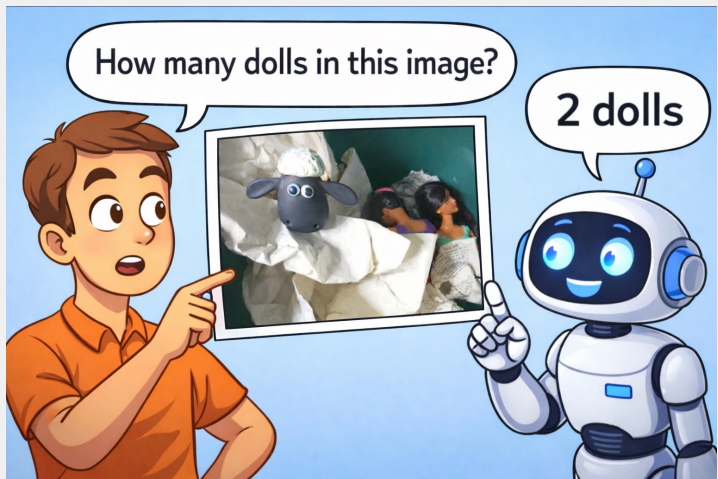


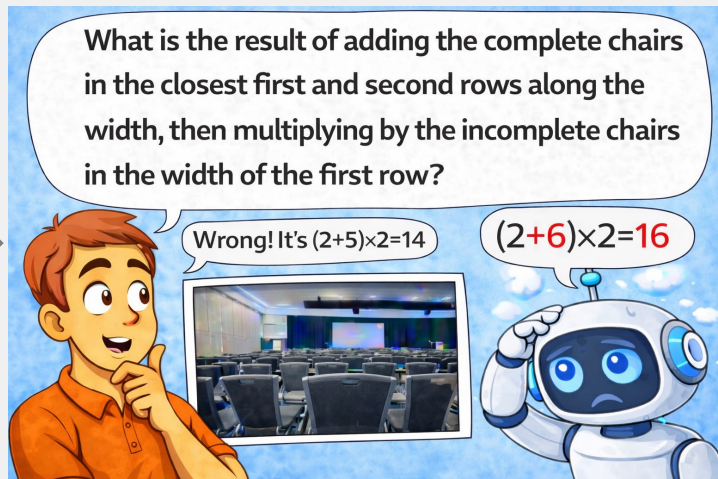
(a) HopChain Framework for Multi-Hop Data Synthesis



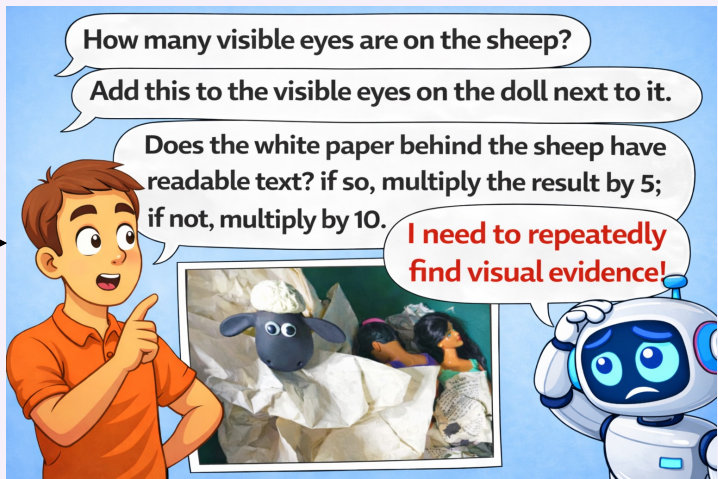
(b) Typical training data: lacks complex reasoning chains relying on visual evidence throughout.



Evaluated with hard problems: minor errors in steps compound into final mistakes.



(c) Synthesized multi-hop data: enforces dependency-linked hops with repeated visual grounding.



Evaluated with hard problems: reduces error cascades and yields more reliable final answers.

